OPIM 201 Operations Management

PROCESS ANALYSIS
(C&T, Ch. 3)

Learning Objective

- Basic Process Analysis
    - Understand the concept of process capacity and resource utilization
    - Identify process bottleneck
- Analyze processes with one product/demand type, as well as with multiple product/demand types.

The Goal of an Enterprise

Maximize Shareholder Value → *Increase Earnings* → *Sales – Operating Expenses – Capital Expenses*

THROUGHPUT (RATE): *The rate at which the system generates revenue/sales*

- *Production is not Revenue; Revenue = Min {Production, Demand}*
- *Capacity Utilization is not the Goal, only a possible means to achieve the goal*

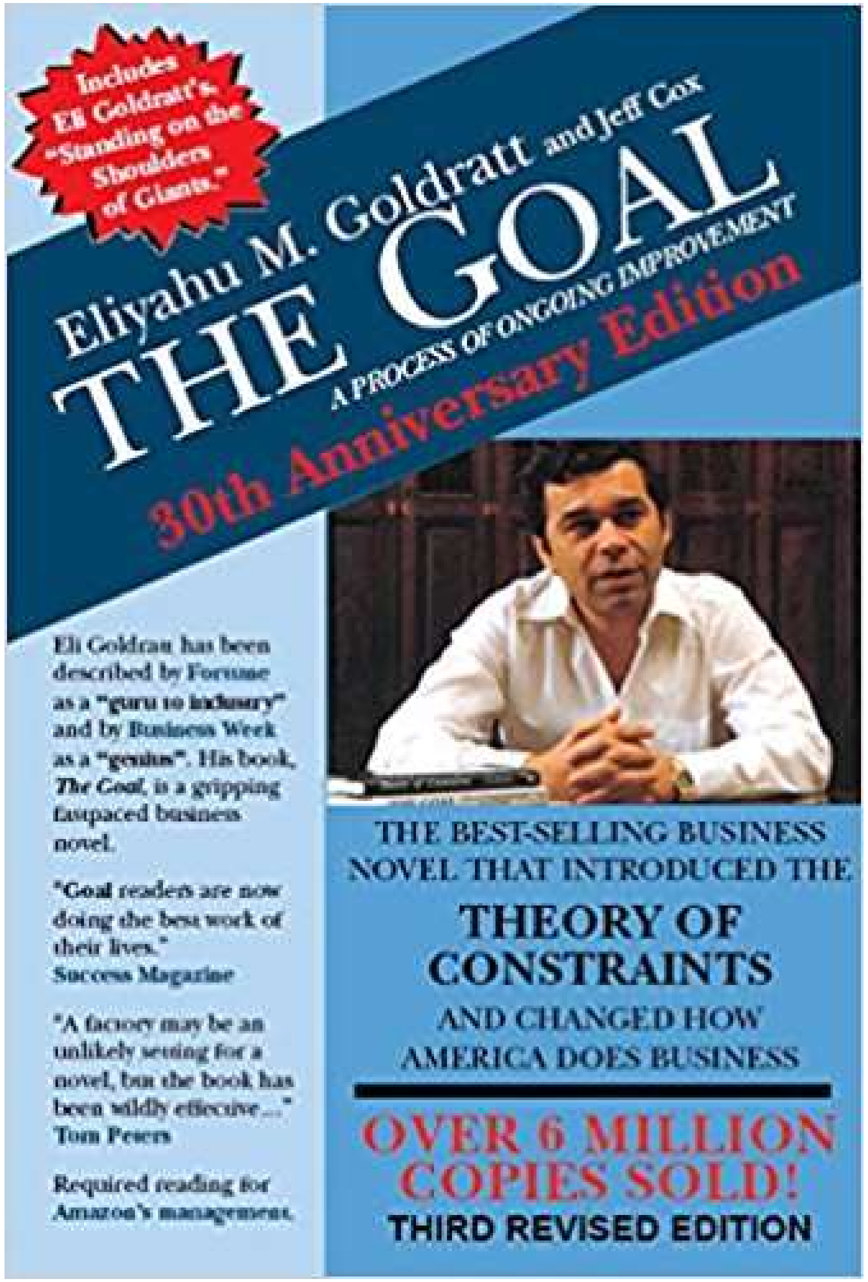
OPERATING EXPENSES: *The rate at which the system generates costs*

INVENTORY: *The level of capital invested in system*

- *It costs money to make money… just don't take too much*
- *Money costs money*
  - *Debt: Interest, shareholder risk increases*
  - *Equity: Dilution of shareholding*

*$$ Going into the System = **Operating Expense*** → *$$ Sitting in the System = **Inventory*** → *$$ Coming out of the System = **Throughput***

The Goal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"Standing on the
Shoulders
of Giants."
Eliyahu M. Goldratt and Jeff Cox
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A PROCESS OF ONGOING IMPROVEMENT
30th Anniversary Edition
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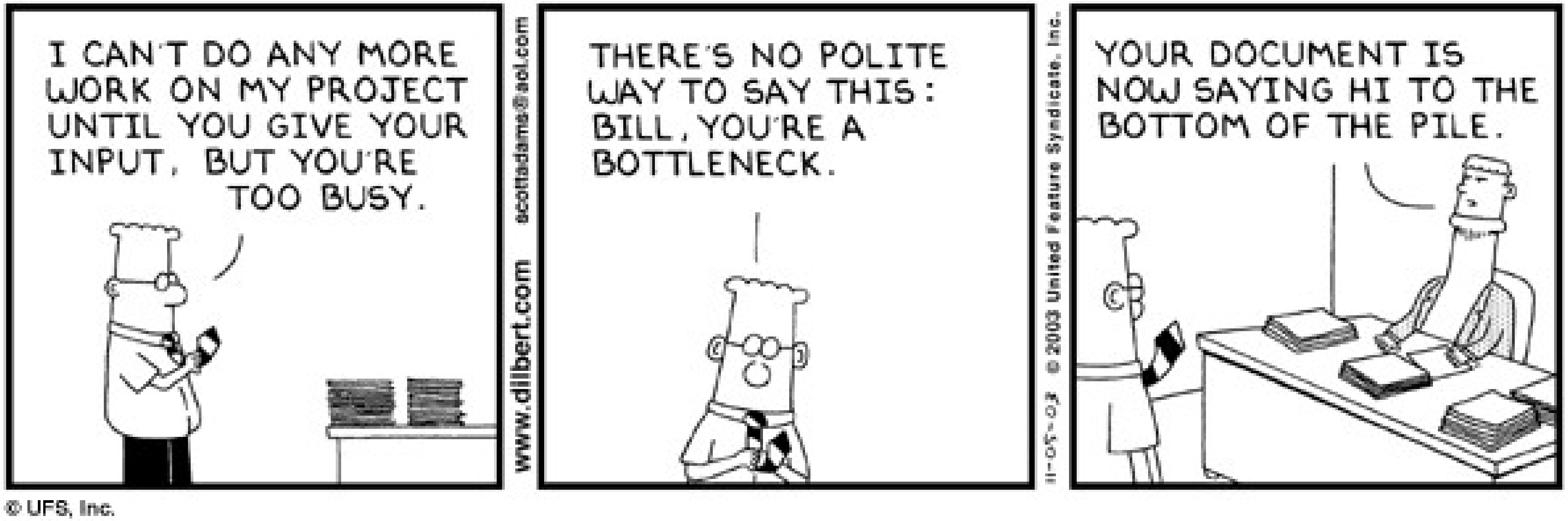
Motivation

Suppose you manage an assembly operation for a firm that produces kick scooters. The assembly process is divided between three workers (resources). You are interested in the performance of this operation. For example:

- What is the maximum daily production of scooters? (concerns revenue.)
- What are the utilization of the resources (e.g., workers, machinery)? How can you improve the utilization? What determines the utilization of these resources? (concerns costs.)
- A customer wants to place an order for 100 kick scooters and wants to know the lead time, i.e., how long you would take to deliver the scooters. What lead time should you quote? (concerns customer service.)
  - You want to quote an accurate and reliable lead time (fair not only to the customer but also to you)!

Managing Bottlenecks



Symbols Used for Process Analysis

- Activity: ▭
- Flow: ⟶
- Storage: ▽ or △

Summary of Process Vocabulary

- **Processing Time**: The time it takes a resource (e.g. worker) to complete one flow flow. (Sometimes, processing time is also referred as cycle time.)
- **Capacity**: The maximum number of flow units that can be completed by the resource per unit of time; Capacity = $\frac{1}{\text{Processing Time}}$.
  - If there are m workers at the activity, Capacity = $\frac{m}{\text{Processing Time}}$;

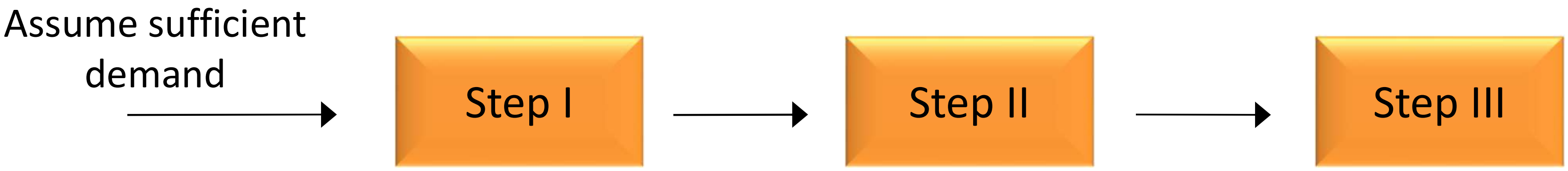
    Sum of capacities of all workers

    i.e., (Effective) Processing Time at the activity = $\frac{\text{Processing Time}}{m}$.
- **Utilization**: The fraction of time the resource is active, $\frac{\text{Busy Time}}{\text{Available Time}}$.
  - Utilization = $\frac{\text{Flow rate}}{\text{Available Capacity}} \leftarrow \frac{\text{what actually gets done}}{\text{what theoretically can get done}}$
- **Bottleneck**: Resource with the highest utilization.
  - It is often the resource with the lowest capacity or largest processing time.
- **System (Process) Capacity**: The maximum number of flow units that can be completed by a process per unit of time.
  - Very often, System Capacity = Bottleneck Capacity.

Summary of Process Vocabulary

- **Flow Rate, R** = Minimum {System Capacity, Demand}.
  - Process can be capacity-constrained or demand-constrained.
- **(System) Cycle Time, CT:** The time between completions of successive units.
  - CT = $^{1}/_{\mathrm{R}}$
- **Break–Even Quantity** $= \dfrac{\textbf{Fixed Costs}}{\textbf{Sales Price per Unit – Variable Cost Per Unit}}$

What is the System Capacity/Throughput?



	Step I	Step II	Step III
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity			
Bottleneck			
Bottleneck Capacity			
System Capacity			
Utilization			
Benefits of Additional Resources			

Where should additional resources be allocated?
How should the work be re-allocated?

Basic Process Vocabulary

- **Processing Time**: The time it takes a resource (e.g. worker) to complete one flow unit. (Sometimes, processing time is also referred as cycle time.)
- **Capacity**: The maximum number of flow units that can be completed by the resource per unit of time; Capacity = $\frac{1}{\text{Processing Time}}$.
- **Utilization**: The fraction of time the resource is active, $\frac{\text{Busy Time}}{\text{Available Time}}$.
  - Utilization = $\frac{\text{Flow rate}}{\text{Available Capacity}} \leftarrow \frac{\text{what actually gets done}}{\text{what theoretically can get done}}$

Motivating Example

Suppose a barber can complete a haircut in exactly 10 minutes. Then his processing time is 10 minutes per haircut and his capacity is 1/10 haircuts per minute or 6 (= 1/10 $\times$ 60) haircuts per hour.

Identifying the Bottleneck

Assume sufficient demand → 20 → Step I → 20 → Step II → 20 → Step III

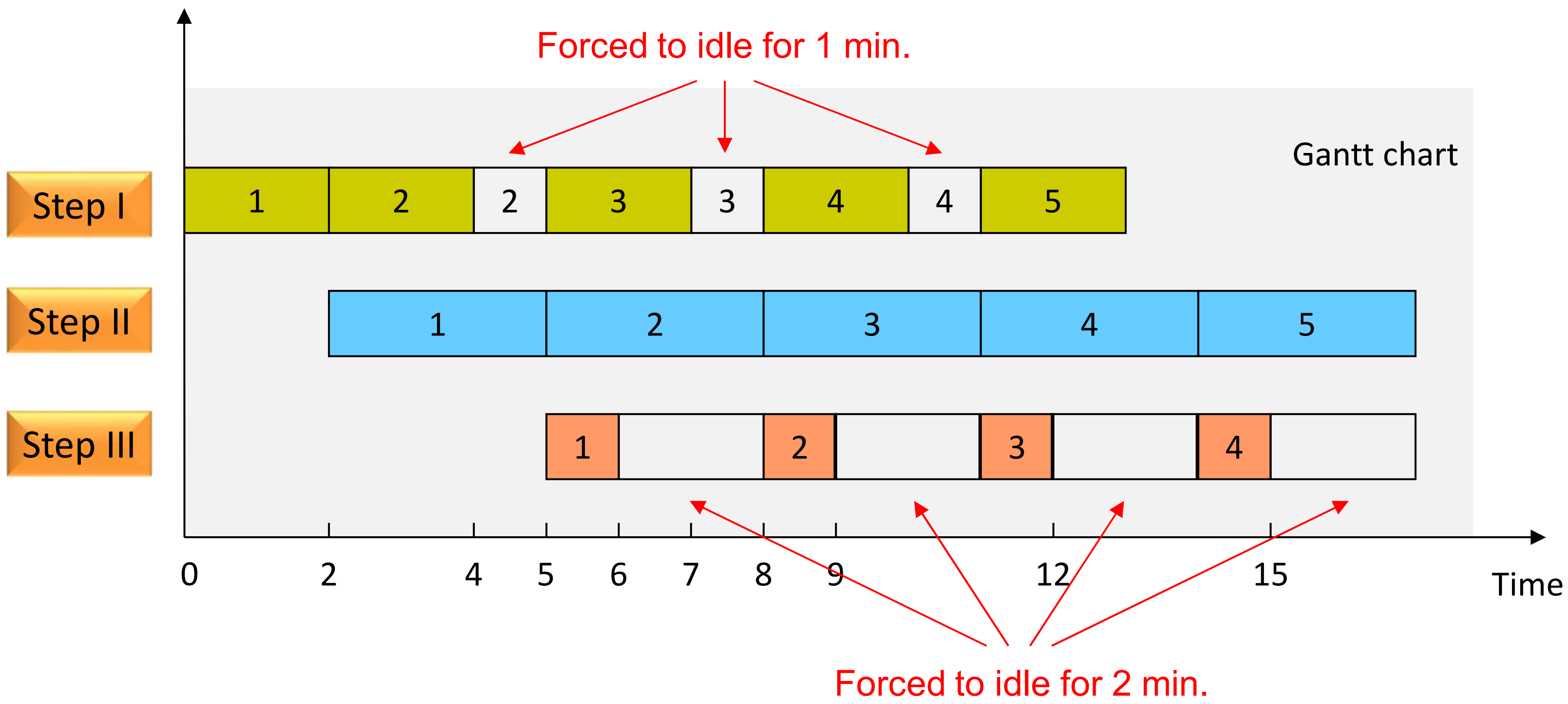
	Step I	Step II	Step III
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity	½ units/min or 30 units/hr	20 units/hr	60 units/hr
Bottleneck	Step II		
Bottleneck Capacity	20 units/hr		
System Capacity	20 units/hr		
Utilization	20/30 ≈ 66.67%	20/20 = 100%	20/60 ≈ 33.33%
Benefits of Additional Resources	0	+	0

Bottleneck = Step with Lowest Capacity

System Capacity = Bottleneck Capacity

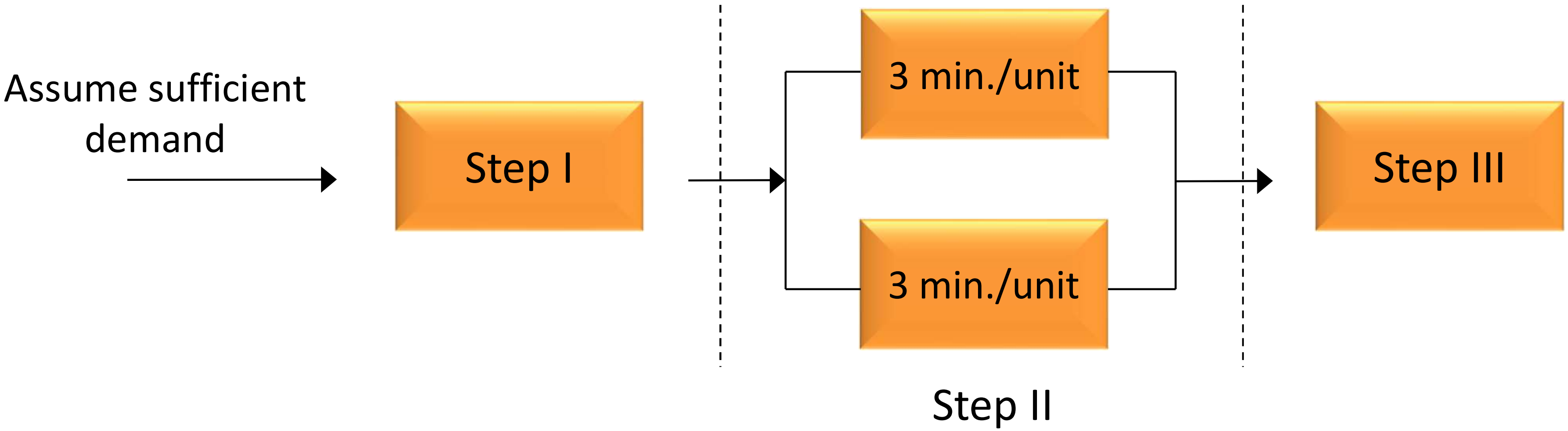
Additional resources should be allocated at the bottleneck step.
If possible, work should be re-allocated from the bottleneck step to other steps.

The Weakest Link: Why the Bottleneck Determines System Capacity



Alleviating Bottleneck …

- Suppose we decide to add a second resource (processing time of 3 minutes per unit) to Step II (current bottleneck). What is the resulting capacity of step II? What is the capacity of the system?



Processing Time	2 minutes/unit	?	1 minute/unit
Capacity	½ units/min.	?	1 unit/min.
Bottleneck			
Bottleneck Capacity			
System Capacity			

Basic Process Vocabulary

- **Processing Time**: The time it takes <u>a resource</u> (e.g. worker) to complete one unit flow. (Sometimes, processing time is also referred as <u>cycle time</u>.)
- **Capacity**: The maximum number of flow units that can be completed by the resource per unit of time; Capacity = $\frac{1}{\text{Processing Time}}$.
  - If there are m workers at the activity, Capacity = $\frac{m}{\text{Processing Time}}$;

    i.e., (Effective) Processing Time at the activity = $\frac{\text{Processing Time}}{m}$.

Sum of capacities of all workers

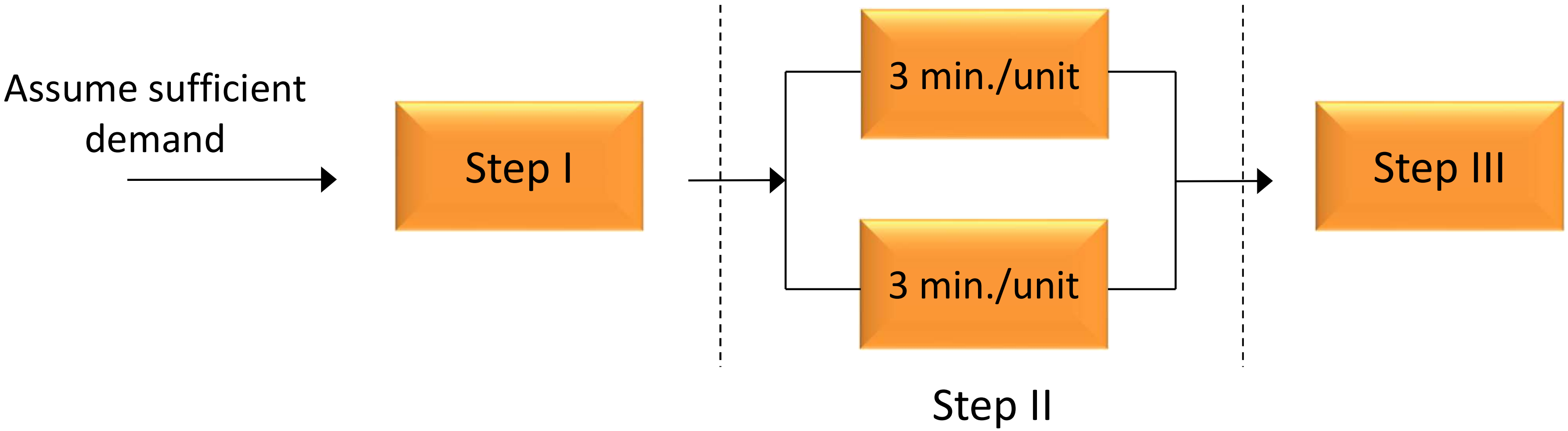
<u>Motivating Example</u>

Suppose a barber can complete a haircut in exactly 10 minutes. Then his processing time is 10 minutes per haircut and his capacity is 1/10 haircuts per minute or 6 (= 1/10 $\times$ 60) haircuts per hour.

If there are two barbers, each capable of completing a haircut in exactly 10 minutes. Then the effective capacity of the barber shop is 1/5 (= 2$\times$1/10) haircuts per minute of 12 haircuts per hour.

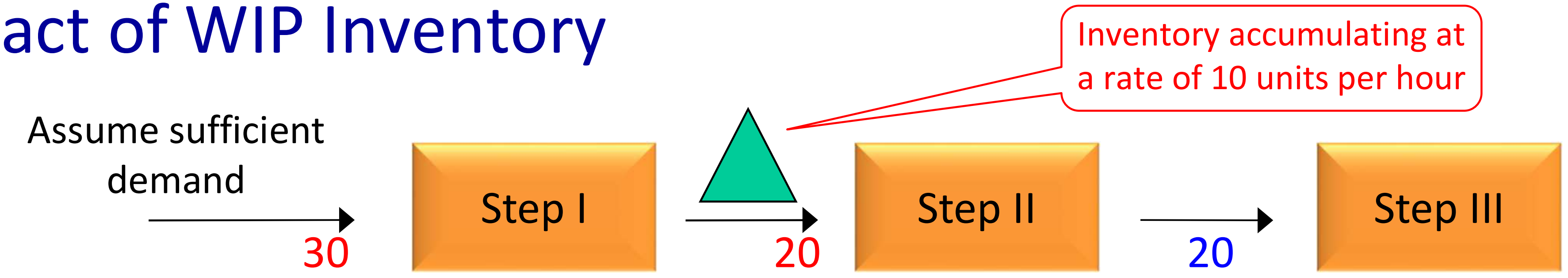
Alleviating Bottleneck …

- Suppose we decide to add a second resource (processing time of 3 minutes per unit) to Step II (current bottleneck). What is the resulting capacity of step II? What is the capacity of the system?



Processing Time	2 minutes/unit	3/2 = 1.5 minutes/unit	1 minute/unit
Capacity	½ units/min.	(1/3)×2 = 2/3 units/min.	1 unit/min.
Bottleneck	Step I (new bottleneck!)		
Bottleneck Capacity	½ units/min = 30 units/hr		
System Capacity	30 units/hr		

Impact of WIP Inventory



Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity	30 units/hr	20 units/hr	60 units/hr
Bottleneck	Step II		
Bottleneck Capacity	20 units/hr		
System Capacity	20 units/hr		
Utilization	30/30 = 100%	20/20 = 100%	20/60 ≈ 33.33%

Does WIP inventory improve the system performance?

Quoting a Lead Time – System Cycle Time (CT)

- Suppose a customer wants to place an order for 1000 units. What lead time should you quote the customer?
  - How long does it take to deliver 1000 units?
    - If you start the system empty, then the first unit takes 6 minutes
    - The remaining units?

→ Step I → Step II → Step III

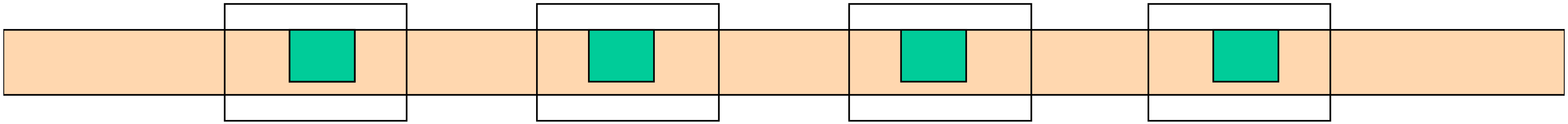
	Step I	Step II	Step III
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity	30 units/hr	20 units/hr	60 units/hr

Recall: Cycle Time (CT) versus Flow Time (T)

- **Cycle Time (CT)**: The time between completions of successive units.
  - **CT =** $\dfrac{\mathbf{1}}{\textbf{Flow Rate}}$

Example Assembly Line

A moving conveyor that passes a series of 4 workstations in a uniform time interval of 10 minutes. What are the cycle time & flow time?



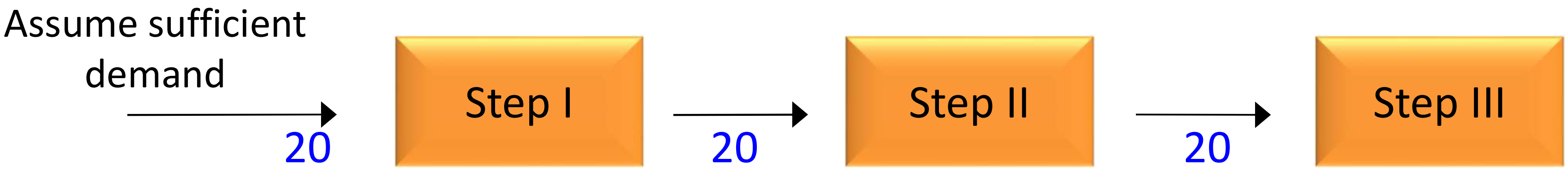
CT = time between successive units coming off the end of line

= 10 minutes (Flow Rate = 1/10 units/min or 6 units/hr)

T = time a unit spent in the system

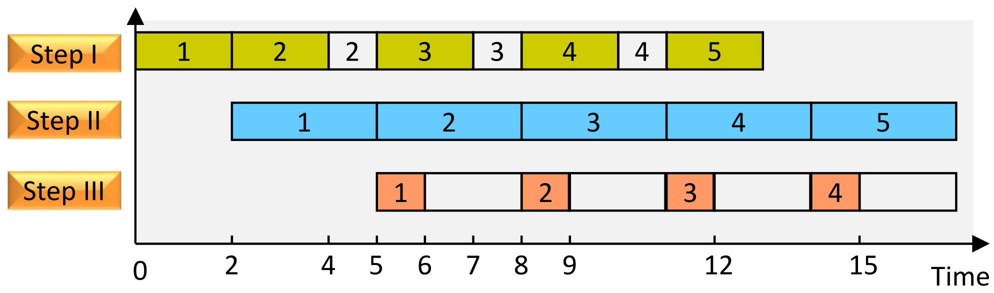
= 40 minutes

(System) Cycle Time



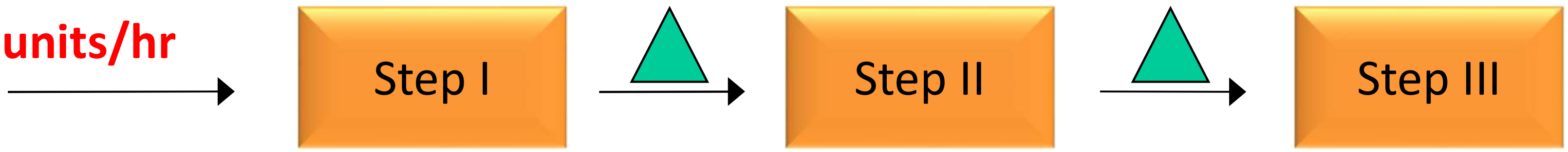
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity	30 units/hr	20 units/hr	60 units/hr
Bottleneck	Step II		
System Capacity	20 units/hr		
Cycle Time	3 minutes (= (1/20 hr)×(60 min. per hr))		
Utilization	20/30 ≈ 66.67%	20/20 = 100%	20/60 ≈ 33.33%

- If starts system empty, it takes (6+3×999) = 3003 min. (≈ 50hrs.) to deliver 1000 units.
- If system is already running, it takes (3×1000) = 3000 min. (= 50 hrs.) to deliver 1000 units.
- If you operate 10 hrs per day, then quote 5 days.



Demand-Constrained Process

Demand
= 10 units/hr

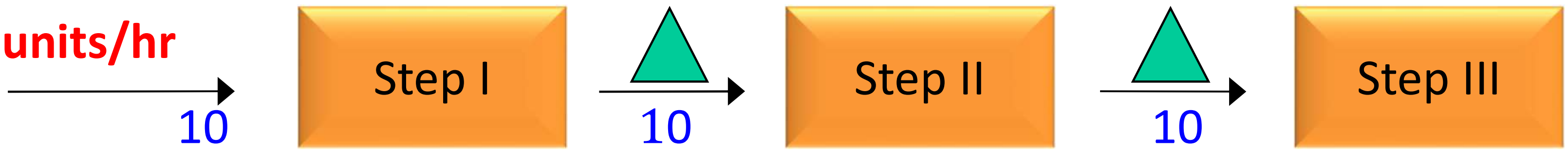


	Step I	Step II	Step III
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity			
Bottleneck			
System Capacity			
Flow Rate (R)			
Cycle Time (CT)			
Utilization			

Suppose demand is 10 units per hour. Assess the performance of the process. Does the bottleneck change?

Demand-Constrained Process (cont'd)

Demand
= 10 units/hr



Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity	30 units/hr	20 units/hr	60 units/hr
Bottleneck	Step II		
System Capacity	20 units/hr		
Flow Rate (R)	10 units/hr	***Flow Rate = Min. {Demand, System Capacity}***	
Cycle Time (CT)	1/10 hr or 6 minutes	***CT = 1/R***	
Utilization (Flow Rate/Capacity)	10/30 ≈ 33.33%	10/20 = 50%	10/60 ≈ 16.67%

*A process can be **demand-constrained** (the process is not at maximum capacity; R = demand) or **capacity-constrained** (the process is at maximum capacity; R = system capacity).*

Discuss: The Impact of Quality or Yield Loss Problems

Input or inflow
= 30 units/hr

Step I

Quality
Control

60%

40%

Step II

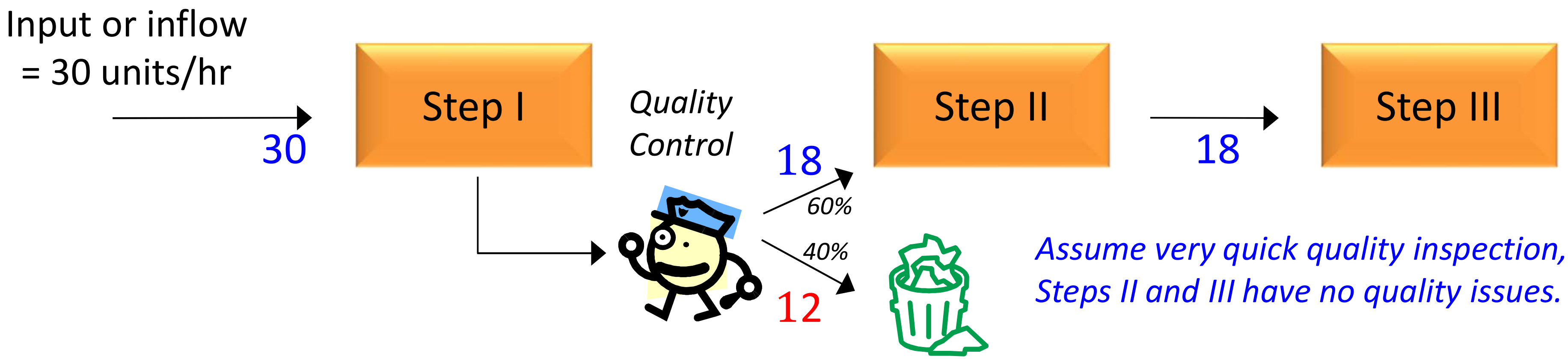
Step III

Assume very quick quality inspection, Steps II and III have no quality issues.

	Step I	Step II	Step III
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity			
Inflow or Workload			
System Capacity			
Utilization at Maximum Capacity			
Bottleneck ?			

What is the system capacity? Which is the bottleneck?

The Impact of Quality or Yield Loss Problems



	Step I	Step II	Step III
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity	30 units/hr	20 units/hr	60 units/hr
Inflow or Workload	30 units/hr	18 units/hr ($=60\% \times 30$)	18 units/hr
System Capacity	18 units/hr		
Utilization at Maximum Capacity	30/30 = 100%	18/20 = 90%	18/60 = 30%
Bottleneck ?	Step I		

- *Step I becomes the (new) bottleneck; system capacity reduces to 18 “good” units/hr.*
- *Quality problem induces “Loss Capacity” at the original bottleneck, Step II.*

Managing Bottleneck

- Bottlenecks Drive System Performance
  - Utilization of a non-bottleneck resource is not determined by its own potential but by the bottleneck
  - Lost Capacity at bottleneck is lost capacity for whole system (e.g., machine breakdown or yield loss)
- Finding The Bottleneck: Symptoms
  - The Resource with Minimum Capacity (or Largest Processing Time)
  - The Resource which is always busy (or with Highest Utilization)
  - The Resource with longest wait
  - The Resource with most inventory
- Improving Bottleneck Performance
  - Investments in improving bottleneck capacity improve system capacity – improving non-bottleneck performance is a mirage
  - Always utilize the bottleneck fully – 100% utilization of a non-bottleneck is wasteful

Theory of Constraints

- **IDENTIFY** the system constraints.
- Decide how to **EXPLOIT** the system constraints.
- **SUBORDINATE** everything else to that decision.
- **ELEVATE** the system constraints.
- If, in the previous steps, the constraints have been broken, **REPEAT** Step 1.

Tesla Found This Methodology Useful!!

Matt Robinson
3 years ago

News

A Production 'Bottleneck' Means Tesla Is Way Behind On Model 3 Deliveries

Tesla delivered just 220 Model 3s in the third quarter of 2017, well behind Elon Musk's original target

REMIND ME LATER

A few months ago, Tesla CEO Elon Musk laid down some **bold production targets** for the new Model 3. Production is supposed to be hitting 20,000 units per month from December, with 100 built in July and 1500 in August, but so far, Tesla has delivered...220.

Case Studies and Examples

Theory of Constraints Examples

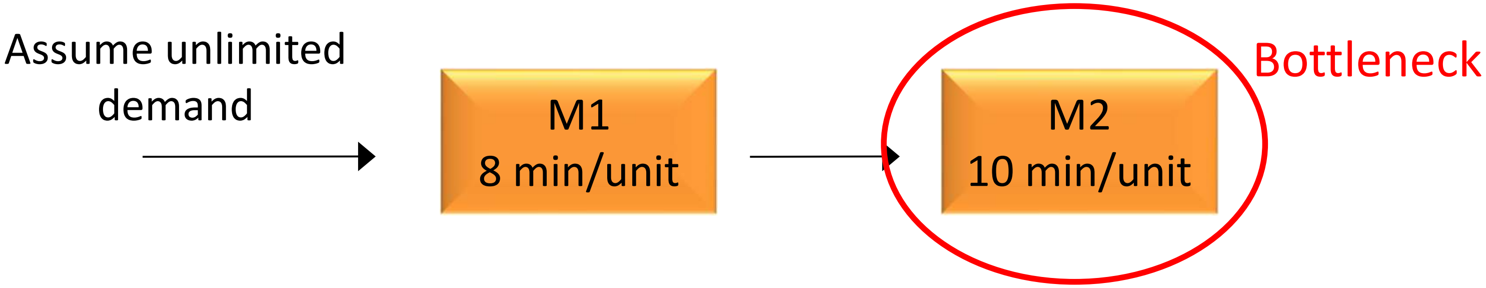
Implementation Examples

Over 5,000 organizations around the world have achieved breakthrough results with the Theory of Constraints, including SMEs, regional corporates and even large multinational such as:

JCB
Dr.Reddy's
Lilly
TANISHQ
Cadillac
mazda
P&G Procter & Gamble
Ford
GE
intel
Pratt & Whitney A United Technologies Company
Samsonite
Texas Instruments
ADITYA BIRLA GROUP
croma The Electronics Megastore
EMBRAER
Godrej
LARSEN & TOUBRO It's all about Imagineering
Lufthansa Technik
TATA STEEL

Check Your Understanding: Example 1

A process involves two sequential tasks done at M1 & M2 (e.g. mixing & baking) with processing time 8 min. & 10 min. resp. Assume unlimited demand. What are the system cycle time and flow rate? What is the average utilization of the two machines?



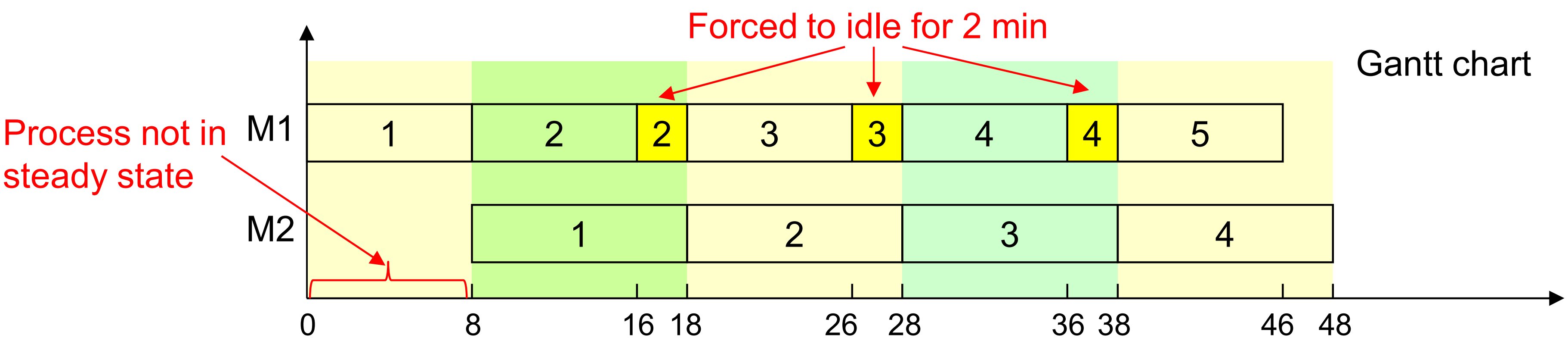
System Cycle Time = 10 min

Flow Rate = 1/10 units/min (capacity-constrained)

Utilization of M1 = (1/10)/(1/8) = 80%

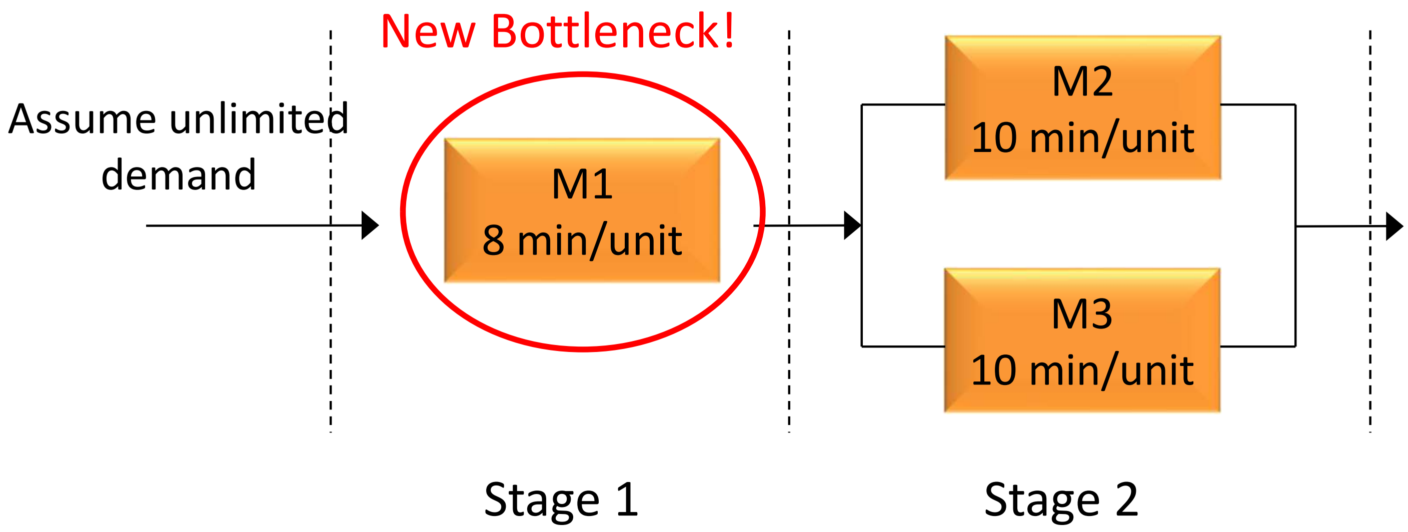
Utilization of M2 = (1/10)/(1/10) = 100%

Average Utilization = (0.8+1)/2 = 90%



Check Your Understanding: Example 2

Suppose we add a second machine to stage 2 in Example 1; i.e., stage 2 has parallel machines {M2, M3}. What are the capacity and effective processing time at stage 2? What is the flow rate?



Capacity at Stage 2 = $(1/10)\times2$ = 1/5 units/min

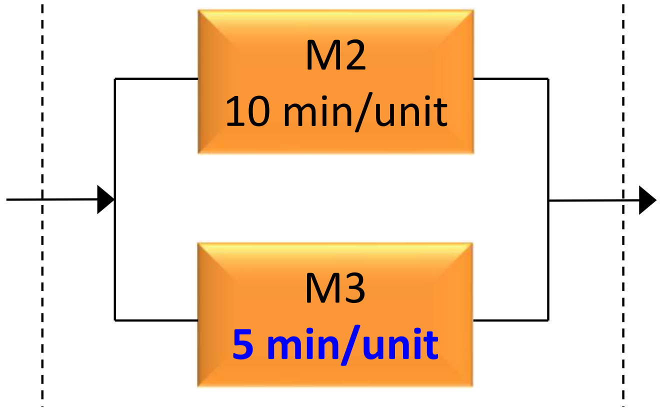
<u>Effective</u> processing time at Stage 2 = 10/2 = 5 min

System CT = 8 min

R = 1/8 units/min

Check Your Understanding – Non-Identical Parallel Machines

Suppose M3 in stage 2 of Example 3 has a processing time of 5 minutes instead 10 minutes. What is the effective capacity at stage 2?



Capacity at M2 = 1/10 per minute

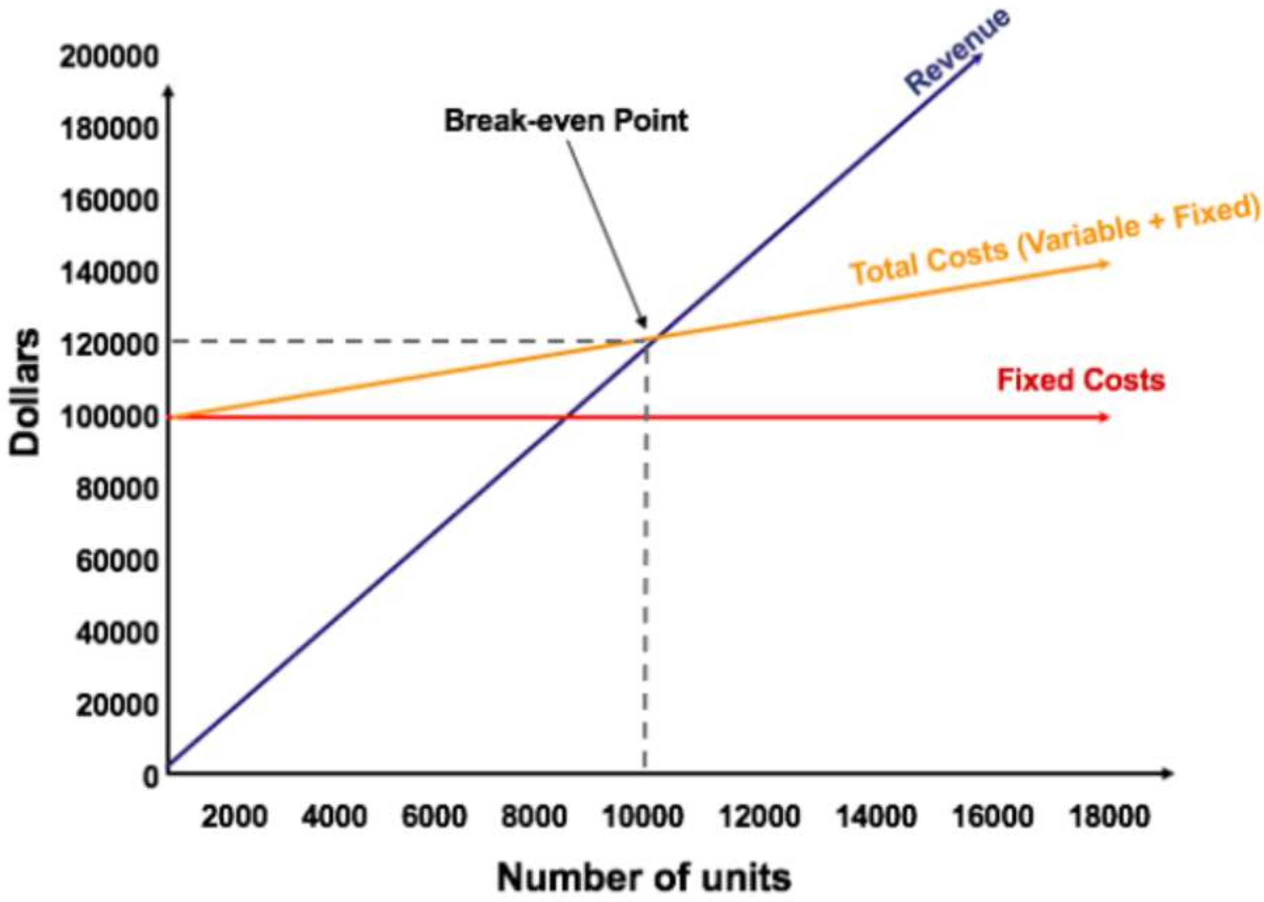
Capacity at M3 = 1/5 per minute

Effective capacity at stage 2 = 1/10 + 1/5

= 3/10 per minute (or 3 units every 10 minutes)

Break-Even Analysis

- Break-even analysis in economics, business, and cost accounting refers to the point at which total costs and total revenue are equal; it is used to determine the number of units or amount of revenue needed to cover total costs (fixed and variable costs).



Source: CFI, “Break Even Analysis”, https://corporatefinanceinstitute.com/resources/accounting/break-even-analysis/

Break-Even Analysis: Applications

- By identifying the exact number of units or amount of revenue needed to cover costs, businesses can use break-even analysis to make informed decisions about investment, pricing, cost management, and sales strategies. For example, break-even analysis is useful in the following:
  - Introducing new products and services, operational expansion or marketing promotion
    - Assessing the viability of these activities.
  - Assessing Viability of Investment
    - Determine the break-even point (quantity or time) for recoupment of initial investment or start-up costs so as to secure funding and show investors the plan for your business.
  - Annual Operating Planning (AOP) or Budget Balancing
    - Identify the break-even quantity that would cover the total annual costs.

Break-Even Analysis: Computations

- The formula for break-even analysis is as follows:

$$\textbf{Break−Even Quantity} = \frac{\textbf{Fixed Costs}}{\textbf{Sales Price per Unit – Variable Cost Per Unit}}$$

where:

- **Fixed Costs** are costs that do not change with varying output (e.g., salary, rent, building, machinery)
- **Sales Price per Unit** is the selling price per unit
- **Variable Cost per Unit** is the variable cost incurred to create a unit

- Derivation:

$$\text{Total Revenue} = \text{Total Costs} \quad (= \text{Fixed costs} + \text{Variable costs})$$

$$\text{Break−Even Quantity} \times \text{Unit Sales Price} = \text{Fixed Costs} + \text{Break−Even Quantity} \times \text{Unit Variable Cost}$$

$$\text{Break−Even Quantity} \times (\text{Unit Sales Price} - \text{Unit Variable Cost}) = \text{Fixed Costs}$$

Subway – Firm Level Information

Started as Pete's Super Submarines in Connecticut in 1965

Now, largest sandwich chain with 34,000+ stores in 98 countries

Estimated revenue: $12 Billion (compared to McDonald's at $23 Billion)

Franchise model – each restaurant is independently owned and operated

Goal: "to become the number one Quick Service Restaurant in the World"

Average revenue per store: $445k (compared to $2.3M at McDonald's)

Subway – The Franchisee's Perspective

- Relatively inexpensive to open a new store:
  - Start-up costs for a restaurant are $100k to $200k
  - No cooking, no grills, and no fryolators
  - Potentially very small stores (as little as 600sqft is possible)
  - Compares to about $1 Million to open a McDonald's
- Franchise model
  - 8 percent of revenue as royalty fee
  - 4.5 percent of revenue as a marketing fee
  - Initial Franchise fee of $15k
- Detailed training and instructions provided by franchiser (Doctor's Associates)
  - Two week training course
  - Detailed operations manual

Subway – Assembly Line for Sandwiches

- Suppose you run a Subway outlet that is currently set up with three stations plus a toasting machine* (see figure). The start-up cost is $150k and the initial franchise fee is $15k. You have also signed a two-year lease at an annual rental of $102,000.
- You hire one worker at an hourly wage of $12 to manage each station. You operate 8 hours per day, 365 days per year.
- The average ingredient cost per sandwich sold is $2 and the average unit selling price is $7.

	Task	Seconds	3+ Support
Station 1	Greet Customer	4	1
	Take Order	5	
	Get Bread/Wrap/Salad Bowl	4	
	Cut Bread	3	
	Meat	12	
	Cheese	9	
	Toasting	30	Places Pulls
Station 2	Onions	3	2
	Lettuce	3	
	Tomatoes	4	
	Cucumbers	5	
	Pickles	4	
	Green Peppers	4	
	Black Olives	3	
	Hot Peppers	2	
	Place Condiments	5	
	Wrap/Napkins & Bag	13	
Station 3	Offer Fresh Value Meal™	3	3
	Offer Cookies	14	
	Ring on Register	20	
	Total Seconds	120 - 150	
Support Position	Phone/Fax Orders		4
	Baking Bread/Cookies		
	Filling In on Line		
	Restocking Sandwich Unit		
	Cleaning Customer Area		

Questions

1) Suppose the average demand is 90 sandwiches per hour. What is the (direct) labor cost per sandwich sold?
2) To facilitate annual operating planning, determine the break-even quantity for your annual operations, assuming that the start-up cost and the initial franchise fee are fully depreciated or amortized over a span of 5 years.

Subway – Assembly Line for Sandwiches (cont'd)

Task	Description	Time(sec)
A	Greet Customer	4
B	Take Order	5
C	Get Bread / Wraps / Salad Bowl	4
D	Cut Bread	3
E	Meat	12
F	Cheese	9
G	Toasting	30
H	Onions	3
I	Lettuce	3
J	Tomatoes	4

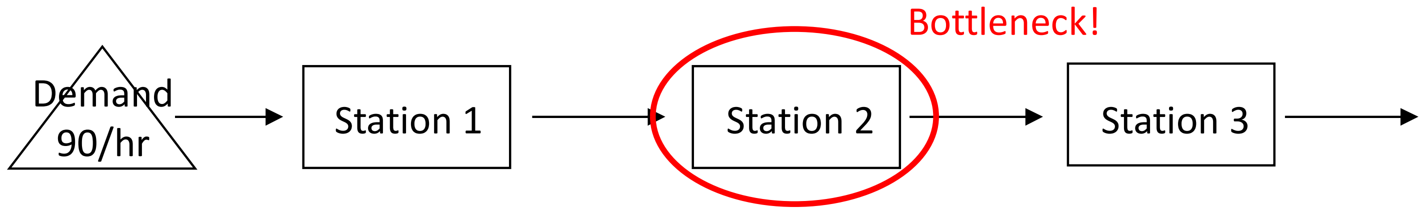
Task	Description	Time(sec)
K	Cucumbers	5
L	Pickles	4
M	Green Peppers	4
N	Black Olives	3
O	Hot Peppers	2
P	Place Condiments	5
Q	Wrap/Napkins & Bag	13
R	Offer Fresh Value Meal	3
S	Offer Cookies	14
T	Ring on Register	20

Current Set-up:

A, B, C, D, E, F	G	H, I, J, K, L, M, N, O, P, Q	R, S, T
37	**30**	**46**	**37**

Question 1 (Hourly Demand: 90 Sandwiches)

Ignore Task G, Toasting (oven)



	Worker 1	**Worker 2**	**Worker 3**
Activity Time	37 seconds/unit	46 seconds/unit	37 seconds/unit
Capacity	1/37 unit per second = 97.3 units/hour	1/46 unit per second = 78.3 units/hour	1/37 unit per second =97.3 units/hour
Process Capacity	Minimum {97.3, 78.3, 97.3 units/hour} =78.3 units/hour		
Flow rate	Demand = 90 units/hour (Target CT = $^{60\times60}/_{90}$ = 40 seconds) Flow rate = Minimum {demand, process capacity} = 78.3 units/hour		
Utilization	$^{78.3}/_{97.3} = 80.5\%$	$^{78.3}/_{78.3} = 100\%$	$^{78.3}/_{97.3} = 80.5\%$

The process is _capacity_-constrained. (System CT (= 46 sec.) > Target CT (= 40 sec.))

Avg. labor utilization = $\frac{80.5+100+80.5}{3} \approx 87\%$

Labor cost per sandwich = $\frac{3 \text{ workers} \times \$12\ /\ hr}{78.3 \text{ units/hr}}$ = $\$0.46/\text{unit}$

Question 2 (Break-Even Analysis for Annual Operating Plan)

Item	Category	Cost
Start-up Costs	Fixed	$150,000/5 = $30,000
Franchise Fee	Fixed	$15,000/5 = $3,000
Rental	Fixed	$102,000
Wages	Fixed	3×$12/hr×8×365 = $105,120
Royalty Fee	Variable	8%×$7 = $0.56
Marketing Fee	Variable	4.5%×$7 = $0.315
Ingredient Costs	Variable	$2

Question 2 (Break-Even Analysis for Annual Operating Plan)

- Total Annual Fixed Costs

  = \$30,000+\$3,000+\$102,000+\$105,120

  = \$240,120

- Unit Variable costs

  = \$2+ (8%+4.5%)×\$7

  = \$2.875

- **Break Even Quantity = \$240,120 / (\$7 – \$2.875) ≈ 58,211 units per year**

- The hourly break-even quantity is 58,211/(8 × 365) ≈ 20 units, which is much less than the current hourly sales quantity of 78.3 units (although hourly demand could be less than 78.3 units due to demand variability).

Remarks

- Note that labor costs (and in general, salaries or wages) are typically fixed costs (or sunk costs in the short run). Nonetheless, it is still useful to compute labor cost per unit sold so as to analyze and identify ways to improve the cost structure and the bottom line of a company.
- For this course, we only consider the case where the start-up cost and initial franchise fee are fully depreciated or amortized evenly over a span of 5 years. In practice, we have to incorporate income tax and net present values so that the analysis becomes more complicated. For example, these sunk costs cannot be depreciated evenly over the 5 years, but rather depreciated following some non-linear formula that is based on net present values taking into consideration the resulting taxable income.

Recall …

Assume sufficient demand $\xrightarrow{20}$ Step I $\xrightarrow{20}$ Step II $\xrightarrow{20}$ Step III

	Step I	Step II	Step III
Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Capacity	½ units/min or 30 units/hr	20 units/hr	60 units/hr
Bottleneck	Step II		
Bottleneck Capacity	20 units/hr		
System Capacity	20 units/hr		
Utilization	20/30 ≈ 66.67%	20/20 = 100%	20/60 ≈ 33.33%

Bottleneck = Step with Lowest Capacity

System Capacity = Bottleneck Capacity

Process Analysis Extensions: Multi-Product System

- Products X and Y. **X is 80% of the product mix, Y is 20% of the product mix.**

Assume sufficient demand → Step I → Step II → Step III

	Step I	Step II	Step III
X Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Y Processing Time	3 minutes/unit	2 minutes/unit	1 minute/unit
Effective Processing Time			
Capacity			
Bottleneck			
Bottleneck Capacity			
System Capacity			

What is the capacity of this system?

Process Analysis Extensions: Multi-Product System

- Products X and Y. **X is 80% of the product mix, Y is 20% of the product mix.**

Assume sufficient demand → Step I → Step II → Step III

	Step I	Step II	Step III
X Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Y Processing Time	3 minutes/unit	2 minutes/unit	1 minute/unit
Effective Processing Time	$2\times\mathbf{0.8} + 3\times\mathbf{0.2}$ $= 2.22$ min	2.8 min	1 min
Capacity	$1/2.22 \times 60$ ≈ 27.27 units/hr	21.43 units/hr	60 units/hr
Bottleneck	Step II		
Bottleneck Capacity	21.43 units/hr		
System Capacity	21.43 units/hr		

Process Analysis Extensions: Multi-Product System

- Products X and Y. **X is 40% of the product mix, Y is 60% of the product mix.**

Assume sufficient demand → Step I → Step II → Step III

	Step I	Step II	Step III
X Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Y Processing Time	3 minutes/unit	2 minutes/unit	1 minute/unit
Effective Processing Time			
Capacity			
Bottleneck			
Bottleneck Capacity			
System Capacity			

What is the capacity of this system?

Process Analysis Extensions: Multi-Product System

- Products X and Y. **X is 40% of the product mix, Y is 60% of the product mix.**

Assume sufficient demand → Step I → Step II → Step III

	Step I	Step II	Step III
X Processing Time	2 minutes/unit	3 minutes/unit	1 minute/unit
Y Processing Time	3 minutes/unit	2 minutes/unit	1 minute/unit
Effective Processing Time	$2\times\mathbf{0.4} + 3\times\mathbf{0.6}$ = 2.6 min	2.4 min	1 min
Capacity	$1/2.6 \times 60$ ≈ 23 units/hr	25 units/hr	60 units/hr
Bottleneck	Step I		
Bottleneck Capacity	23 units/hr		
System Capacity	23 units/hr		


LEMON-LIME
Sprite
ZERO SUGAR
Coca-Cola
zero
SUGAR
ZERO CALORIE COLA
CHERRY
Coca-Cola
ORANGE VANILLA
Coca-Cola
CAFFEINE FREE
Fanta
ORANGE
CAFFEINE FREE
Fanta
GRAPE
Ginger Ale
SOFT DRINK
Schweppes
200 ml
Mojito
MINT & LIME
FLAVOURED DRINK
Schweppes


ESPRESSO
Coca-Cola
WITH
COFFEE
GREAT COKE TASTE
BLENDED WITH RICH
COFFEE BEANS
NO SUGAR

Key Lessons

- In a process with a series of tasks:
  - Bottleneck drives system performance – it determines the maximum flow rate through the process.
  - System capacity can only be increased by improving the capacity at the bottleneck, after which the bottleneck may move to another resource.
  - Focus on utilizing the bottleneck fully!

- In a process with multiple products, the mix of demand types may influence the bottleneck and the system performance.